

MTH203: Sequences and series

Lecture 15

1. USING THE RATIO TEST AND ROOT TEST TO PROVE DIVERGENCE

In the previous lecture, we saw the Root Test and the Ratio Test for convergence of series. These ideas can also be used to prove divergence of series. We first look at the Ratio Test.

Theorem 1 (Ratio Test for Divergence). *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. Suppose there exists an integer N and a real number $r > 1$ such that for every $n > N$, we have $|a_{n+1}/a_n| \geq r$. Then, the series $\sum_{n=1}^{\infty} a_n$ is divergent.*

Proof. Suppose the series $\sum_{n=1}^{\infty} a_n$ is convergent. Then, (by Lecture 12, Theorem 1) the sequence a_n must converge to 0. So, the sequence $(|a_n|)_{n \in \mathbb{N}}$ must also converge (see Tutorial 3, Problem 2).

Let N and r be as in the statement of the theorem. Then for $n > N$, we have $|a_{n+1}| \geq |a_n|r > |a_n|$. Thus, the sequence $(|a_{N+k}|)_{k \in \mathbb{N}}$ is a strictly increasing sequence. Note, in particular, that $|a_{N+2}| > |a_{N+1}| \geq 0$.

The sequence $(|a_{N+k}|)_{k \in \mathbb{N}}$ must converge to 0, since it is a subsequence of the sequence $(a_n)_{n \in \mathbb{N}}$ which converges to 0. However, since $(|a_{N+k}|)_{k \in \mathbb{N}}$ is a strictly increasing sequence, it must converge to the supremum of the set $\{|a_{N+k}|\}_{k \in \mathbb{N}}$ (see Lecture 8, Theorem 7). Thus, $0 = \sup\{|a_{N+k}|\}_{k \in \mathbb{N}}$. In particular, $0 \geq |a_{N+2}|$. However, we have observed above that $|a_{N+2}| > 0$. This is a contradiction.

Thus, the series cannot be convergent. This completes the proof. $\square$

This has the following consequence, which is also commonly referred as the Ratio Test for Divergence:

Corollary 2. *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers such that $\lim_{n \rightarrow \infty} |a_{n+1}/a_n|$ exists and is greater than 1. Then the series $\sum_{n=1}^{\infty} a_n$ diverges.*

This is easily derived from the above theorem in the same manner that Corollary 7 of Lecture 14 was derived from Theorem 5 of that lecture.

We now look at the corresponding results for the Root Test.

Theorem 3 (Root Test for Divergence). *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. Suppose there exists an integer N and a real number $r > 1$ such that for every $n > N$, we have $\sqrt[n]{|a_n|} \geq r$. Then, the series $\sum_{n=1}^{\infty} a_n$ is divergent.*

Corollary 4. *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers such that $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ exists and is greater than 1. Then the series $\sum_{n=1}^{\infty} a_n$ diverges.*

We omit the proofs of these two results because they are similar to the arguments given above for the Ratio Test.

2. THE “BOUNDARY CASE” OF THE ROOT TEST AND RATIO TEST

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence such that $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = 1$. In this case, the Ratio Test does not provide us with any conclusion. In fact, in this case, the series could be convergent or divergent. For example, the series $\sum_{n=1}^{\infty} (1/n)$ and $\sum_{n=1}^{\infty} 1/n^2$ both have this property. The first series is divergent while the second is convergent.

Similarly, if $(a_n)_{n \in \mathbb{N}}$ is a sequence such that $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$. In this case, the Root Test does not provide us with any conclusion. Again, the series $\sum_{n=1}^{\infty} (1/n)$ and $\sum_{n=1}^{\infty} 1/n^2$ both have this property.

In these boundary cases, where both these test fail, one has to employ some other techniques to prove convergence or divergence.

3. CAUCHY'S CONDENSATION TEST

Recall that we proved the divergence of the series $\sum_{n=1}^{\infty} (1/n)$ using a special trick. That trick can be generalized and applied to the summation of any non-increasing sequence of positive real numbers.

Theorem 5 (Cauchy's Condensation Test). *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers such that $a_n \geq a_{n+1}$ and $a_n \geq 0$ for all $n \in \mathbb{N}$. Then, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.*

Proof. For any natural number n , let $s_n := \sum_{k=1}^n a_k$ denote the n -th partial sum of the series $\sum_{k=1}^{\infty} a_k$. Similarly, for any non-negative integer n , let $t_n := \sum_{k=0}^n 2^k a_{2^k}$, which is the n -th partial sum of the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$. Since we have $a_n \geq 0$ for all n , both the sequences, $(s_n)_{n \in \mathbb{N}}$ and $(t_n)_{n \in \mathbb{N} \cup \{0\}}$, are non-decreasing. Thus, for either of these two sequences, convergence is equivalent to boundedness.

We first write the series $\sum_{n=1}^{\infty} a_n$ as follows:

$$(a_1) + (a_2 + a_3) + (a_4 + a_5 + a_6 + a_7) + \cdots$$

The sum in the k -th pair of parentheses is $a_{2^{k-1}} + a_{2^{k-1}+1} + \cdots + a_{2^k-1}$. Since the sequence $(a_n)_{n \in \mathbb{N}}$ is non-increasing, we have

$$a_{2^{k-1}} + a_{2^{k-1}+1} + \cdots + a_{2^k-1} \leq a_{2^{k-1}} + \cdots + a_{2^{k-1}} = 2^{k-1} a_{2^{k-1}}.$$

Adding these inequalities for $k = 1, \dots, n$ (where $n \geq 1$), we get

$$s_{2^n-1} \leq a_1 + 2a_2 + \cdots + 2^{n-1} a_{2^{n-1}} = t_{n-1}.$$

Thus, we see that for any natural number n , we have $s_{2^n-1} \leq t_{n-1}$. If the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is convergent, the sequence $(t_n)_{n \in \mathbb{N} \cup \{0\}}$ is convergent. It converges to its supremum (see Lecture 8, Theorem 7). Let $t = \lim_{n \rightarrow \infty} t_n$. Then $t \geq t_{n-1} \geq s_{2^n-1} \geq s_n$ (since $2^n - 1 \geq n$ for all $n \geq 1$). Thus, the sequence $(s_n)_{n \in \mathbb{N}}$ is also bounded and hence converges.

For the converse, we begin with a similar calculation with a slightly different arrangement of parentheses:

$$(a_1) + (a_2) + (a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \cdots$$

Thus, the first set of parentheses just contains a_1 . For $k \geq 2$, the k -th set of parentheses contains

$$a_{2^{k-2}+1} + \cdots + a_{2^{k-1}}.$$

Again, using the fact that $(a_n)_{n \in \mathbb{N}}$ is non-increasing, we see that

$$a_{2^{k-2}+1} + \cdots + a_{2^{k-1}} \geq a_{2^{k-1}} + \cdots + a_{2^{k-1}} = 2^{k-2} a_{2^{k-1}}.$$

Adding these inequalities for $k = 2, \dots, n$ (where $n \geq 2$), we get

$$\begin{aligned} a_2 + a_3 + a_4 + \cdots + a_{2^n-1} &\geq a_2 + 2a_4 + \cdots + 2^{n-2} a_{2^{n-1}} \\ &= (1/2)[2a_2 + 4a_4 + \cdots + 2^{n-1} a_{2^{n-1}}] \\ &= (t_{n-1} - a_1)/2 \end{aligned}$$

Thus, $s_{2^n-1} - a_1 \geq (t_{n-1} - a_1)/2$ for all $n \geq 2$. Thus, $s_{2^n-1} \geq (t_{n-1} + a_1)/2$.

Suppose the series $\sum_{m=1}^{\infty} a_m$ is convergent. Then the sequence $(s_m)_{m \in \mathbb{N}}$ converges. Being a non-decreasing sequence, it converges to its supremum (again, by Lecture 8, Theorem 7). Let $s = \lim_{m \rightarrow \infty} s_m$. Then $s \geq s_{2^n-1} \geq (t_{n-1} + a_1)/2$ and so $t_{n-1} \leq 2s - a_1$ for all n . Thus, the sequence $(t_n)_{n \in \mathbb{N} \cup \{0\}}$ is also bounded and hence converges. $\square$

Example 6. We examine the convergence of the series $\sum_{n=1}^{\infty} 1/n^p$ for $p > 0$. If $0 < p < 1$, we have $1/n^p > 1/n$. Thus, by the Comparison Test, $\sum_{n=1}^{\infty} 1/n^p$ diverges for $0 < p < 1$. Thus, we now focus on the case $p > 1$.

We apply Cauchy's Condensation Test. The series $\sum_{n=1}^{\infty} 1/n^p$ converges if and only if the series $\sum_{k=0}^{\infty} 2^k(1/(2^k)^p)$ converges. We have $2^k(1/(2^k)^p) = 2^{k-kp} = (2^{1-p})^k$. Since $p > 1$, we have $2^{1-p} < 1$ and so the series $\sum_{k=0}^{\infty} 2^k(1/(2^k)^p)$ converges. Thus, the series $\sum_{n=1}^{\infty} 1/n^p$ converges for all $p > 1$.

4. ALTERNATING SERIES TEST

We have stated before that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges. We now prove a general result that implies this.

Theorem 7 (Alternating Series Test). *Let $(a_n)_{n \in \mathbb{N}}$ be a non-increasing sequence of real numbers such that $\lim_{n \rightarrow \infty} a_n = 0$. Then the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges.*

Proof. Since $(a_n)_{n \in \mathbb{N}}$ is a convergent sequence, it is bounded. Since it is a non-increasing sequence, it converges to $\inf\{a_n\}_{n \in \mathbb{N}}$. Thus, $a_n \geq 0$ for all n .

For all $n \in \mathbb{N}$, let $s_n := \sum_{k=1}^n (-1)^{k+1} a_k$ be the n -th partial sum of the series $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$. To motivate the idea behind the proof, let us examine the first few partial sums.

We have the obvious inequality

$$s_1 = a_1 \geq 0.$$

We have $s_2 = a_1 - a_2$. Since $a_2 \leq a_1$, we have $0 \leq a_1 - a_2 \leq a_1$. In other words

$$0 \leq s_2 \leq s_1.$$

Now, $s_3 = s_2 + a_3 \geq s_2$. Since $s_1 - s_2 = a_2 \geq a_3$, we have $s_3 = s_2 + a_3 \leq s_1$. Thus, we have the inequality

$$s_2 \leq s_3 \leq s_1.$$

Thus, we can see a pattern. The n -th partial sum seems to always lie between s_{n-2} and s_{n-1} . Also, $s_n \geq s_{n-1}$ if n is odd and $s_n \leq s_{n-1}$ if n is even. We will now prove these observations.

First, observe that $s_n = s_{n-1} + (-1)^{n+1} a_n$. Since $a_n \geq 0$, we see that $(-1)^{n+1} a_n \geq 0$ if n is odd and $(-1)^{n+1} a_n \leq 0$ if n is even. Thus, $s_n \leq s_{n-1}$ if n is even and $s_n \geq s_{n-1}$ if n is odd.

For any $n \in \mathbb{N}$, let I_n denote the closed interval with endpoints s_n and s_{n+1} . By our above observations, we see that $s_{2k} \leq s_{2k-1}$ for any natural number k . Thus, $I_{2k-1} = [s_{2k}, s_{2k-1}]$. On the other hand, $s_{2k+1} \geq s_{2k}$ for any natural number k . Thus, $I_{2k} = [s_{2k}, s_{2k+1}]$.

We will show that $I_{n-1} \subset I_{n-2}$ for any $n \geq 3$. (In particular, this implies that $s_n \in I_{n-2}$, as we have anticipated above.) We will have to look at two cases – n is even or n is odd.

Suppose n is even, say $n = 2k$. Then

$$s_{2k} = s_{2k-1} + (-1)^{2k+1} a_{2k} = s_{2k-1} - a_{2k} \leq s_{2k-1}.$$

On the other hand,

$$s_{2k-1} = s_{2k-2} + (-1)^{2k-1+1} a_{2k-1} = s_{2k-2} + a_{2k-1}.$$

Thus,

$$s_{2k-1} - s_{2k-2} = a_{2k-1} \geq a_{2k}$$

and so $s_{2k} = s_{2k-1} - a_{2k} \geq s_{2k-2}$. This shows that $s_{2k} \in [s_{2k-2}, s_{2k-1}] = I_{n-2}$. Thus,

$$I_{n-1} = [s_{2k}, s_{2k+1}] \subset [s_{2k-2}, s_{2k-1}] = I_{n-2}.$$

Suppose n is odd, say $n = 2k + 1$ (where $k \geq 1$). Then

$$s_{2k+1} = s_{2k} + (-1)^{2k+2} a_{2k+1} = s_{2k} + a_{2k+1} \geq s_{2k}.$$

On the other hand,

$$s_{2k} = s_{2k-1} + (-1)^{2k+1} a_{2k} = s_{2k-1} - a_{2k}.$$

Thus,

$$s_{2k-1} - s_{2k} = a_{2k} \geq a_{2k+1}$$

and so $s_{2k+1} = s_{2k} + a_{2k+1} \leq s_{2k-1}$. This shows that $s_{2k+1} \in [s_{2k}, s_{2k-1}] = I_{n-2}$. Thus,

$$I_{n-1} = [s_{2k}, s_{2k+1}] \subset [s_{2k}, s_{2k-1}] = I_{n-2}.$$

Thus, we have proved that $I_{n-1} \subset I_{n-2}$ for all $n \geq 3$. Thus, $I_{n+1} \subset I_n$ for all $n \geq 1$. In other words, the $\{I_n\}_{n \in \mathbb{N}}$ is a sequence of closed, nested intervals. Thus, by the Nested Interval Property, we deduce that $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Let s be an element of $\bigcap_{n=1}^{\infty} I_n$. Then $s \in I_n$. Since s_n and s_{n+1} are the endpoints of I_n , we have $|s - s_n| \leq |s_n - s_{n+1}|$. So,

$$|s - s_n| \leq |(-1)^{n+2} a_{n+1}| = a_{n+1}.$$

Recall that $\lim_{n \rightarrow \infty} a_n = 0$. Let $\epsilon > 0$ be given. Then, there exists N such that for all $n > N$, we have $|a_n| = |a_n - 0| < \epsilon$. For any such n , we also have $n + 1 > N$ and so $|s - s_n| \leq a_{n+1} < \epsilon$. Thus, $\lim_{n \rightarrow \infty} s_n = s$.

(Note that we picked s to be any element in $\bigcap_{n=1}^{\infty} I_n$. But our conclusion shows that this element is actually unique since we know that a sequence can converge to at most one point.) $\square$

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015.